



By Nitesh Singh

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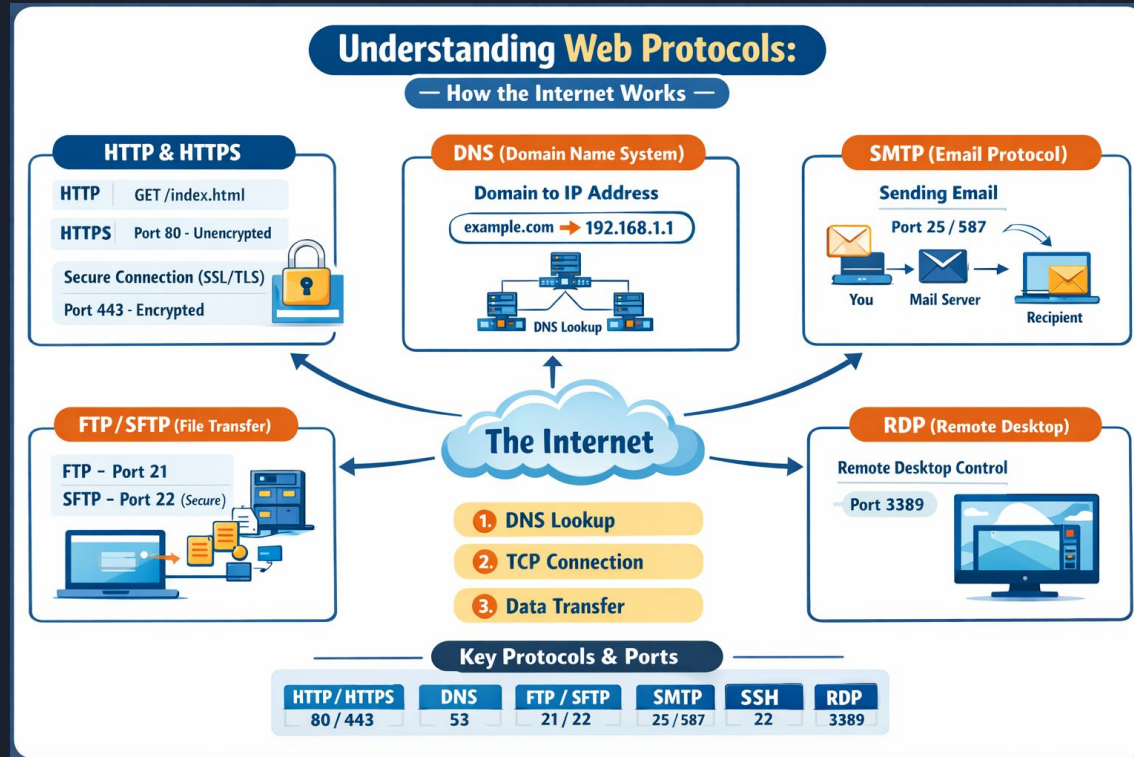


What are Web Protocols ?

A **web protocol** is simply a **set of rules** that define how data is sent and received over a network (especially the internet).

- Humans → language (English, Hindi)
- Computers → protocols (HTTP, HTTPS, DNS, etc.)

Without protocols, devices wouldn't understand each other.





Where Do Web Protocols Fit?

Web protocols operate across layers of the OSI Model:

Layer	Example Protocols
Application	HTTP, HTTPS, DNS, FTP
Transport	TCP, UDP
Network	IP
Data Link	Ethernet, WiFi

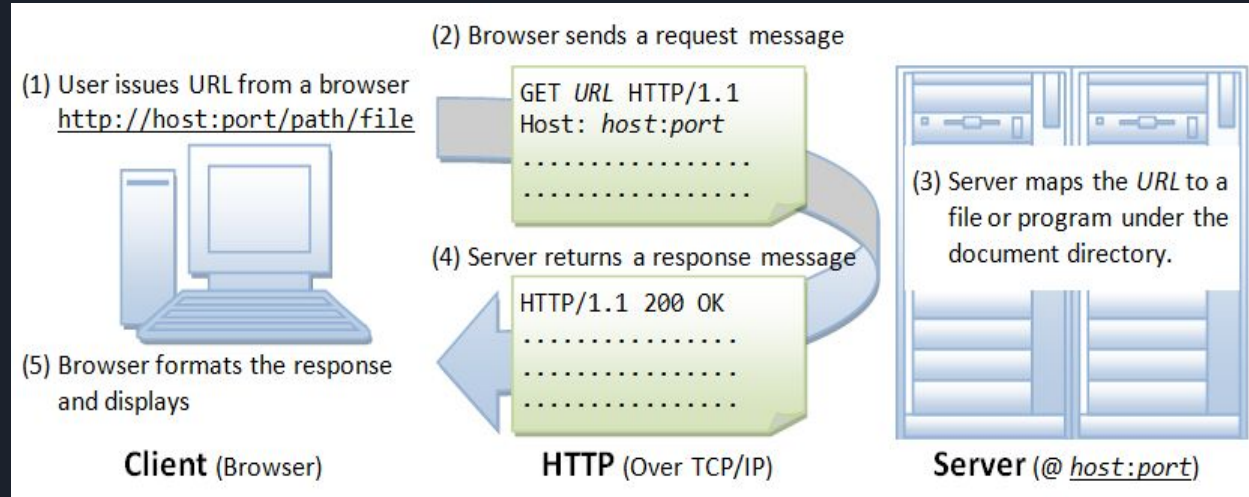


Common Protocols : HTTP - Hypertext Transfer Protocol

What is HTTP?

HTTP is the **core web protocol** used to transfer web pages.

- Works on **Port 80**
- Stateless (IMPORTANT concept)





Common Protocols : HTTP - Hypertext Transfer Protocol

How HTTP Works (Step-by-Step)

Step 1: Client Request

Browser sends request:

GET /index.html HTTP/1.1

Host: example.com

User-Agent: Chrome

Step 2: Server Response

HTTP/1.1 200 OK

Content-Type: text/html

<html>...</html>

Client → Request → Server
Server → Response → Client

SOC Indicators :

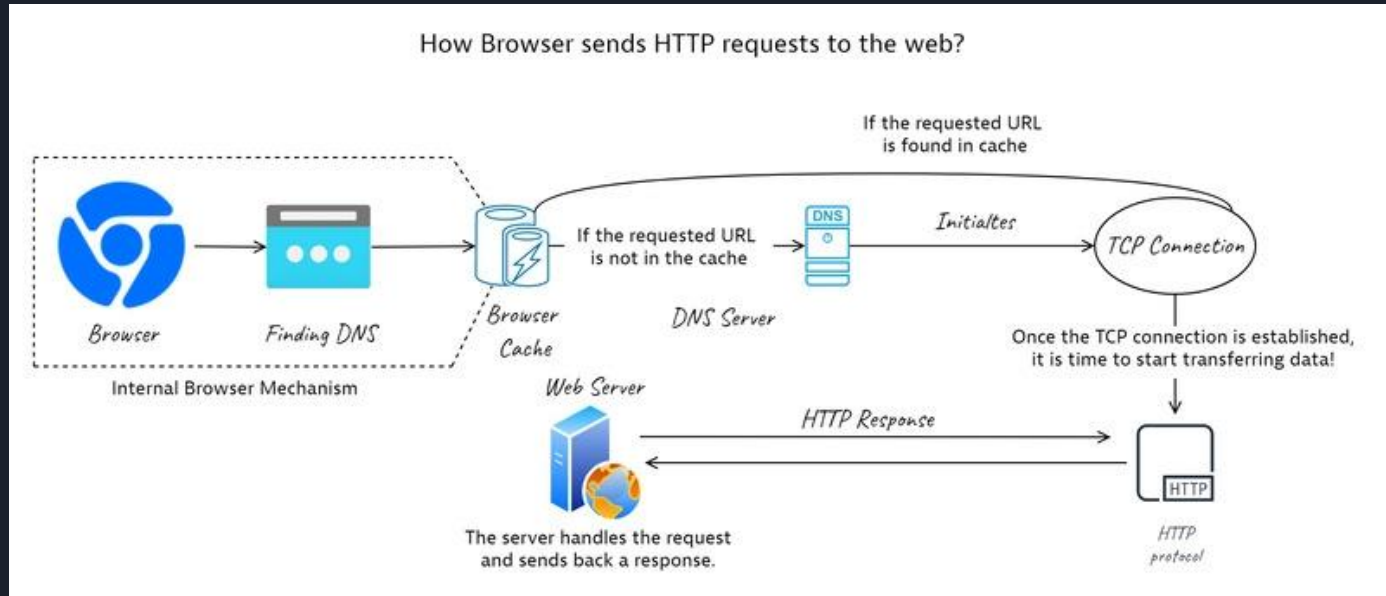
- Suspicious URLs
- Repeated requests (/login brute force)
- Unusual user-agents
- Large POST requests

Attacks

- XSS
- SQL Injection
- Credential harvesting
- HTTP Flood (DDoS)



Common Protocols : HTTP - How Browser Send Requests





HTTP Methods

HTTP defines a set of **request methods** to indicate the purpose of the request and what is expected if the request is successful.

Stateless Nature (VERY IMPORTANT)

HTTP does NOT remember:

- Who you are
- Previous requests

That's why we use:

- Cookies
- Sessions
- Tokens

Method	Purpose
GET	Fetch data
POST	Send data
PUT	Update resource
DELETE	Remove resource
PATCH	Partial update



HTTP Status Codes

HTTP response status codes indicate whether a specific HTTP request has been successfully completed. **Responses are grouped in five classes:**

1. Informational responses (100 – 199)
2. Successful responses (200 – 299)
3. Redirection messages (300 – 399)
4. Client error responses (400 – 499)
5. Server error responses (500 – 599)

Code	Meaning
200	OK
301	Redirect
400	Bad request
401	Unauthorized
403	Forbidden
404	Not found
500	Server error



HTTPS (Secure HTTP)

What is HTTPS?

HTTP + Encryption (via SSL/TLS)

- Works on **Port 443**
- Provides:
 - Confidentiality
 - Integrity
 - Authentication

Why HTTPS Matters

Without HTTPS:

- MITM attacks possible
- Credentials can be sniffed
- Data readable in plaintext



DNS (Domain Name System)

The Domain Name System (DNS) is the phonebook of the Internet. Humans access information online through domain names, like nytimes.com or espn.com.

Web browsers interact through Internet Protocol (IP) addresses. DNS translates domain names to IP addresses so browsers can load Internet resources.

DNS Resolution Process

Converts:

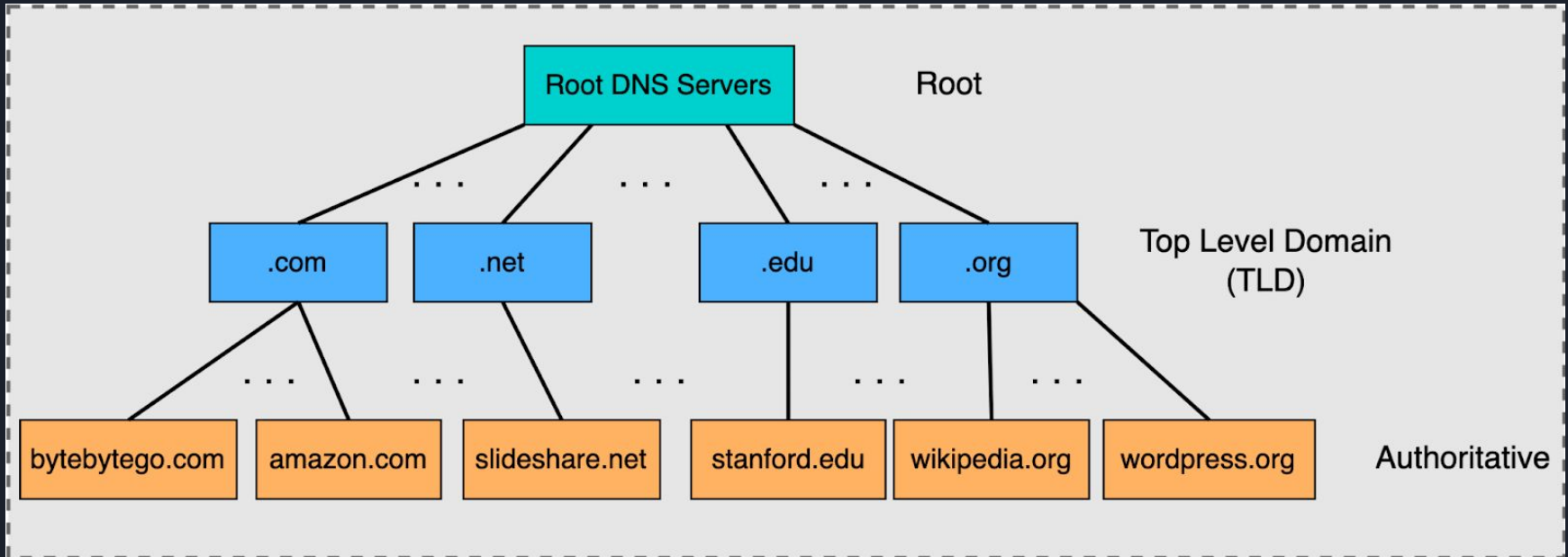
google.com → 142.250.x.x

- Works on **Port 53**
- Uses UDP (mostly)

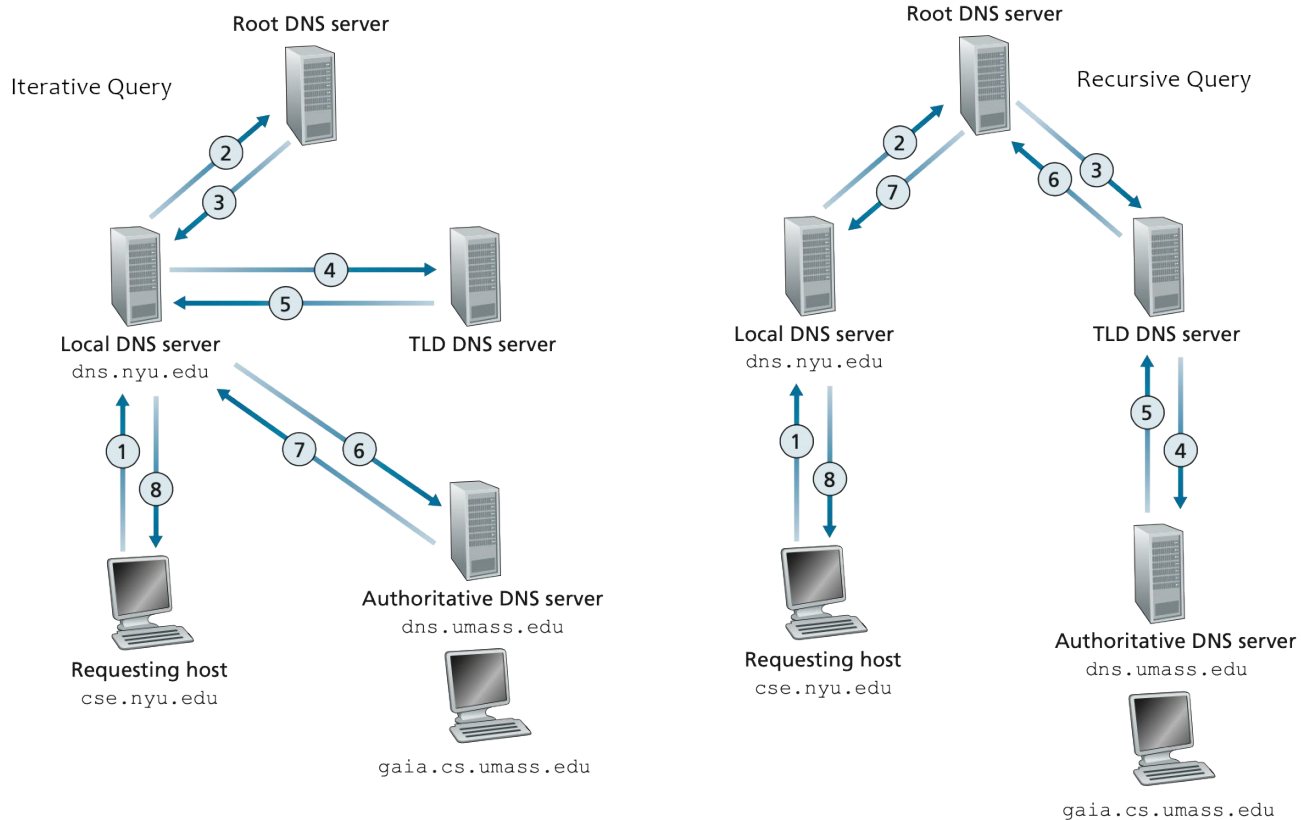
1. Browser cache
2. OS cache
3. Resolver (ISP)
4. Root server
5. TLD server
6. Authoritative server



DNS (Domain Name System)



Protocols





DNS (Domain Name System)

SOC Indicators

- Too many DNS requests
- Suspicious domains
- Random subdomains
- DNS tunneling

Attacks

- DNS Spoofing
- DNS Tunneling (data exfiltration)
- DGA (Domain Generation Algorithm)



FTP / SFTP : Transfers files between systems

FTP (File Transfer Protocol)

- Port 21
- Plaintext (NOT secure)

Modes:

- Active
- Passive

SFTP (Secure FTP)

- Runs over **SSH (Port 22)**
- Encrypted



Protocols

FTP / SFTP : Transfers files between systems

SOC Indicators

- Large file transfers
- Unknown uploads
- Anonymous login

Attacks

- Credential sniffing (FTP)
- Data exfiltration
- Brute force login



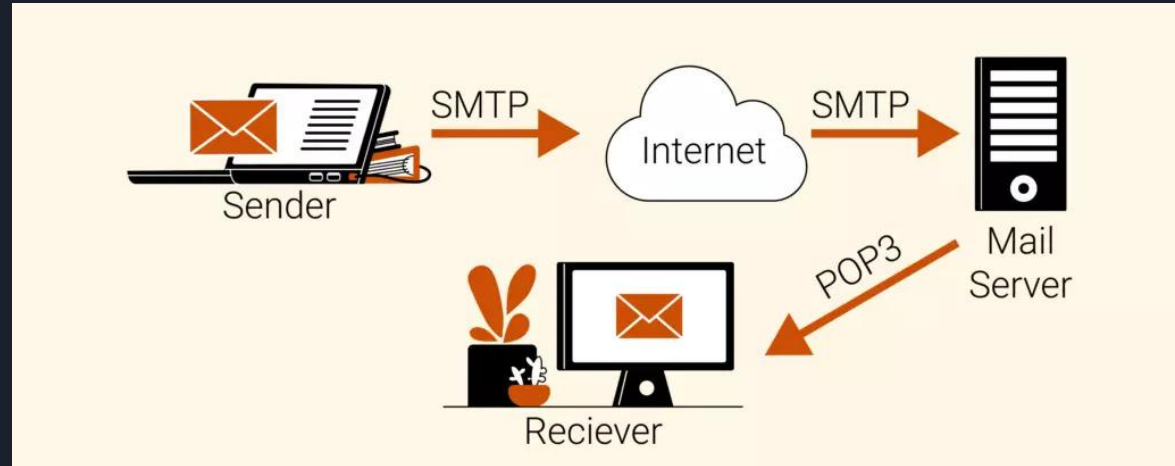
SMTP (Email Sending)

What it does

Sends emails

Port

- 25 (SMTP)
- 587 (secure submission)





SMTP (Email Sending)

SOC Indicators

- Bulk email sending
- Unknown sender domains
- Email spoofing

Attacks

- Phishing
- Spam campaigns
- Email spoofing



SSH (Secure Shell)

What is SSH?

- Secure remote login
- Port 22

SOC Indicators

- Multiple failed logins
- Login from unusual IP
- Root login attempts

Attacks

- Brute force
- Credential theft
- Unauthorized access



RDP (Remote Desktop Protocol)

What it does

Remote control of Windows systems

Port : 3389

SOC Indicators

- Login at odd hours
- Multiple login attempts
- External IP login

Attacks

- RDP brute force
- Lateral movement
- Ransomware entry point



SMB (Server Message Block)

What it does

File sharing in Windows network

Port : 445

SOC Indicators

- Access to admin shares (C\$)
- File movement across systems
- Lateral movement

Attacks

- EternalBlue exploit
- SMB relay
- Lateral movement



What is a Computer Port

A port is a virtual point where network connections start and end. Ports are software-based and managed by a computer's operating system. Each port is associated with a specific process or service.

Ports allow computers to easily differentiate between different kinds of traffic: emails go to a different port than webpages, for instance, even though both reach a computer over the same Internet connection.

There are 65,535 possible port numbers, although not all are in common use.

Computer Ports



- Each port is numbered from 0 to 65535 and is divided into different groups.
- **Well-known Ports (0–1023):** Reserved for standard services like HTTP (80), HTTPS (443), FTP (21), SSH (22).
- **Registered Ports (1024–49151):** Used by specific applications like databases (MySQL 3306, SQL Server 1433).
- **Dynamic/Private Ports (49152–65535):** Temporarily assigned for client connections, like when your browser opens a port to connect to a website.
- Internal Ports only used inside your private network (LAN) for local communication between devices.
- External Ports are open to the internet, routers map them to internal ports so external users can access services safely.

Computer Ports



Important Ports

Port	Protocol	Use
21	FTP	File transfer
22	SSH	Secure remote access
25	SMTP	Email sending
53	DNS	Domain resolution
80	HTTP	Web traffic
443	HTTPS	Secure web
3389	RDP	Remote desktop
445	SMB	File sharing



Real SOC Log Example

Source IP: 185.221.x.x

Destination IP: 192.168.1.10

Port: 3389

Attempts: 500

Status: Failed

Immediately think:

RDP brute force attack